

## Coding & Solution

|                      |              |
|----------------------|--------------|
| <b>Date</b>          | 30 June 2026 |
| <b>Team ID</b>       | XXXXXX       |
| <b>Project Name</b>  | XXXXXX       |
| <b>Maximum Marks</b> | 5 Marks      |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack

<https://github.com/Saajith-07/EduGenie-Google-Gemini-Powered-Learning-Assistant>

2. The solution should meet all functional requirements

3. Include all source files and dependencies (Python 3.10, JavaScript, HTML, CSS)

4. Attach or link to your solution (FastAPI, Uvicorn, Jinja2 template rendering engine)

### Solution Summary

|                                      |                                                                                                                                                                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Field</b>                         | AI Q&A chat; leveled concept explanation (local T5 + Gemini fallback); MCQ quiz generation; text/notes summarization; personalized learning roadmap; JSON-file query/response logging; demo/mock mode; Dockerized deployment         |
| <b>Repository Link / URL</b>         |                                                                                                                                                                                                                                      |
| <b>Programming Language(s)</b>       | User authentication and role-based access; migration from JSON files to a relational database (PostgreSQL); usage analytics dashboard; consolidation of duplicate router/direct API endpoints                                        |
| <b>Framework(s) Used</b>             |                                                                                                                                                                                                                                      |
| <b>Key Features Implemented</b>      | 1) pip install -r requirements.txt<br>2) Copy .env.example to .env and set GEMINI_API_KEY<br>3) uvicorn main:app --reload -> Open http://127.0.0.1:8000<br>4) Docker: docker build -t edugenie . && docker run -p 8000:8000 edugenie |
| <b>Pending / Incomplete Features</b> |                                                                                                                                                                                                                                      |
| <b>Setup / Run Instructions</b>      |                                                                                                                                                                                                                                      |

## Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes |                   |
| 2    | Meaningful variable and function names are used        |                   |
| 3    | Code includes comments / documentation where necessary |                   |
| 4    | Error handling is implemented for critical operations  |                   |
| 5    | The application runs without critical errors           |                   |
| 6    | Code is committed to a version control repository      |                   |

<https://github.com/Saajith-07/EduGenie-Google-Gemini-Powered-Learning-Assistant>

## Additional Notes / Comments

Python 3.10, JavaScript, HTML, CSS

FastAPI, Uvicorn, Jinja2 template rendering engine

AI Q&A chat; leveled concept explanation (local T5 + Gemini fallback); MCQ quiz generation; text/notes summarization; personalized learning roadmap; JSON-file query/response logging; demo/mock mode; Dockerized deployment

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